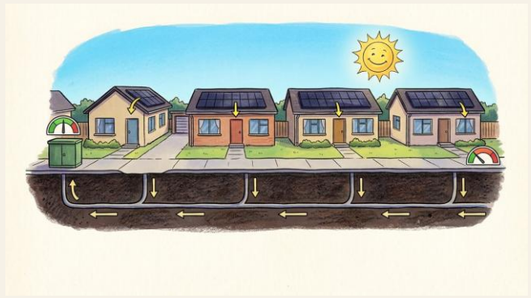
Wrong cable records, unsafe solar limits

Aaquila Konnimera Akbar – Supervised by Archie Chapman and Frederik Geth

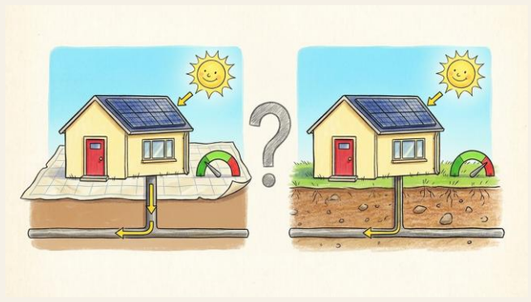
REIT7841 Thesis

Main question: when the network's cable records are wrong, which sets safer solar export limits for homes: a physics-based calculator or AI?

1 The problem



At midday, rooftop solar pushes power back along the cables, so the street voltage rises.



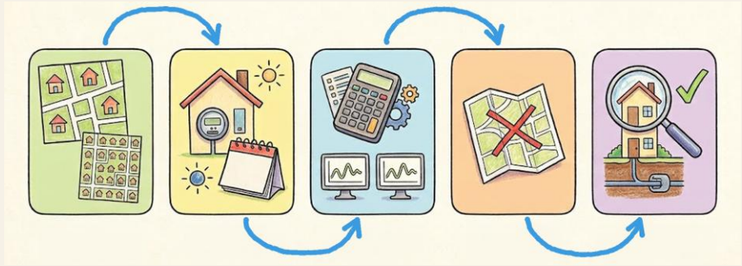
The map says the cable is 16 mm². If it is really a thinner 4 mm², the voltage rises more than the map predicts.

Each home gets an export limit (a Dynamic Operating Envelope, DOE), set so every home stays at or below 253 V.

But cable records are often incomplete, and missing data is filled with the most common cable type [1, 2].

Sub-question (my work so far): how wrong does the physics-based limit get when the cable map is wrong?

2 What we did: method and logic



1 Two realistic CSIRO streets: 12 and 123 homes [3]

2 Home use and sunshine over a day

3 Four-wire calculator, checked by a second program

4 Make the cable map wrong on purpose

5 Test the map's limit on the true street

RECORDED street every service cable 16 mm² → **Same calculator** optimal power flow → **Limit from records** **3.66 kW per home**

TRUE street (test case) 22 segments really 4 mm² → **Same calculator** optimal power flow → **Correct limit** **1.95 kW per home**

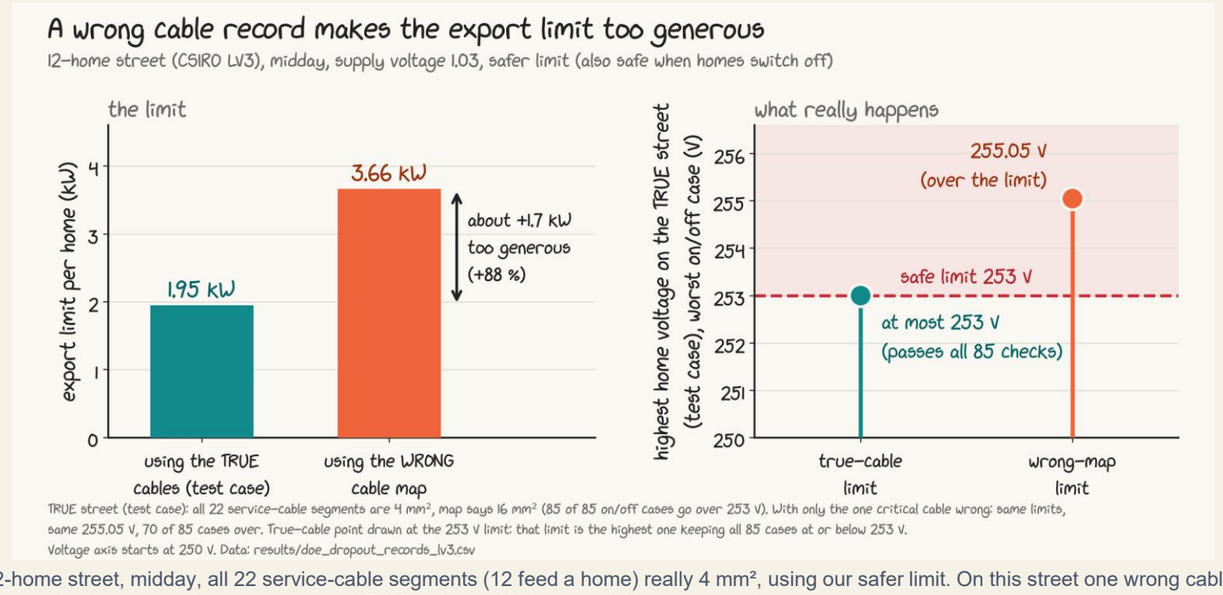
Test: give every home the records' limit on the TRUE street (inverter rules on) and check 253 V. **Gap = 3.664 − 1.947 = 1.718 kW (+88 %).**

Rule: the biggest equal export limit that keeps every home at or below 253 V. Voltage rise ≈ (R×P + X×Q)/V, so a thinner cable (bigger R) means a smaller safe limit.

Errors tested: small scatter (5–20 %), wrong cable type (4 mm², about 3× R), one cable at a time.
Safer limit: also safe when 1 or 2 homes or a phase stop exporting (85 cases). Matches OpenDSS within 0.103 V.

3 What we found

+1.72 kW too much export per home (+88 %) • worst home 255.05 V (safe limit 253 V)



12-home street, midday, all 22 service-cable segments (12 feed a home) really 4 mm², using our safer limit. On this street one wrong cable is enough; the safer limit cuts the gap by 17–25 % here (40 % on the 123-home street).

4 What's next



Build a labelled dataset of streets, records and correct limits, so AI methods can be trained on it.

Then compare them with the physics calculator on the main question.

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Drawings: AI image tool. Code checks AI-assisted (Claude); all results checked by me. Full credits and licences are on GitHub.

References

[1] T. Antić and T. Capuder, Applied Energy, vol. 353, 2024.
[2] M. Mokhtar et al., IEEE Trans. Smart Grid, vol. 11, 2020.
[3] F. Geth et al., CSIRO data, 2025, doi: 10.25919/ghnz-bk28.
Voltage limit: AS 60038:2012. Inverter settings: AS/NZS 4777.2:2020.
Code and results: github.com/aaquila-ka/OperatingEnvelopes

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